

Multiple Choice

1. **Answer:** A

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: A - True, True

2. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: C - True, False

3. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: D - False, True

4. **Answer:** C

Options: A) The use of a non-Gaussian noise model in probabilistic regression.; B) The use of probabilistic modelling for regression.; C) The use of prior distributions on the parameters in a probabilistic model.; D) The use of class priors in Gaussian Discriminant Analysis.

Note: Correct option: C - The use of prior distributions on the parameters in a probabilistic model.

5. **Answer:** C

Options: A) Optimize a convex objective function; B) Can only be trained with stochastic gradient descent; C) Can use a mix of different activation functions; D) None of the above

Note: Correct option: C - Can use a mix of different activation functions

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Step size is too large'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'PCA'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Linear hard-margin SVM.'

Long Form Response

11. **Answer:** `def replace_max_specialchar(text,n): | return (re.sub("[.,]", ":", text, n))`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def find_exponentio(test_tup 1, test_tup 2): | return (res)`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key